

SNF-Bench: Separating Static Drift from Natural Flow in Long-Horizon Fixed-Camera Video Generation

Supplementary Material

001 This supplement records the material supporting the
002 main paper without being required to follow its argu-
003 ment: the region-partition procedure and its sensitiv-
004 ity study, complete audit tables for both tracks at ev-
005 ery horizon, measurement coverage, protocols for the
006 two complementary perceptual checks, and additional
007 qualitative material. The text-conditioned and image-
008 conditioned tracks are reported here at equal depth.
009 Section numbering is independent of the main paper.

010 1. Formal Factor Definitions

011 The definitions below are those referenced from the
012 main paper’s protocol section. Notation follows it: I_t
013 is frame t of a T -frame sequence, Ω_{static} and Ω_{flow} are
014 the region masks, $\mathbf{u}_t$ is the optical flow from I_t to I_{t+1} ,
015 and $\mathcal{E}$ and $\mathcal{L}$ are the early and late windows.

016 Feature-aligned Background Drift. For matched ORB
017 pairs $\mathcal{M}_t$ between the first frame and frame t ,

$$\begin{aligned} D_{\text{feat}}(t) &= \text{median}_{(p,q) \in \mathcal{M}_t} \|p - q\|_2, \\ \text{fBD} &= \frac{100}{d} \cdot \frac{1}{|\mathcal{L}|} \sum_{t \in \mathcal{L}} D_{\text{feat}}(t) \quad (\downarrow), \end{aligned} \quad (1)$$

020 with d the frame diagonal, so fBD is a percentage of
021 image size.

Normalised Background Flow.

$$\text{NBF} = \frac{10^3}{W \Delta t} \cdot \frac{1}{T-1} \sum_{t=1}^{T-1} \frac{1}{|\Omega_{\text{static}}|} \sum_{\mathbf{x} \in \Omega_{\text{static}}} \|\mathbf{u}_t(\mathbf{x})\|_2 \quad (\downarrow), \quad (2)$$

022 with W the frame width and Δt the interval of the
023 sampled pairs actually used for flow estimation, so the
024 quantity is independent of both the native frame rate
025 and the sampling policy.
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027 Global-motion compensation. With $T_t(\mathbf{x}) = s_t R_t \mathbf{x} +$
028 $\mathbf{b}_t$ the robust similarity transform fitted to static-
029 region correspondences and $\mathbf{d}_t(\mathbf{x}) = T_t(\mathbf{x}) - \mathbf{x}$ the dis-
030 placement field it induces,

$$\tilde{\mathbf{u}}_t(\mathbf{x}) = \mathbf{u}_t(\mathbf{x}) - \mathbf{d}_t(\mathbf{x}). \quad (3)$$

Motion-Compensated Foreground Flow and Flow Per-
sistence.

$$\text{MCFF}(\mathcal{W}) = \frac{1}{|\mathcal{W}|} \sum_{t \in \mathcal{W}} \frac{1}{|\Omega_{\text{flow}}|} \sum_{\mathbf{x} \in \Omega_{\text{flow}}} \|\tilde{\mathbf{u}}_t(\mathbf{x})\|_2 \quad (\uparrow), \quad (4)$$

$$\text{FP} = \min \left(\frac{\text{MCFF}(\mathcal{L})}{\text{MCFF}(\mathcal{E}) + \epsilon}, c \right) \quad (\uparrow), \quad (5)$$

clipped at a fixed c so that a vanishing early window
cannot produce an unbounded ratio.

Drift leakage and attenuation. With F_{static} the mean
flow magnitude over Ω_{static} and $F_{\text{raw}}, F_{\text{comp}}$ the mean
raw and compensated magnitudes over Ω_{flow} , all in the
late window,

$$\text{DLR} = \frac{F_{\text{static}}}{F_{\text{raw}} + \epsilon} \quad (\downarrow), \quad \text{DAR} = 1 - \frac{F_{\text{comp}}}{F_{\text{raw}} + \epsilon} \quad (\downarrow). \quad (6)$$

DLR is not a fraction and is deliberately unbounded:
DLR > 1 indicates static support moving more than
the region intended to move. DAR is stored signed
and reported clipped to $[0, 1]$.

047 2. Region Partition

Derivation. For every generated sequence we aver-
age optical-flow magnitude [8] over the leading 12%
of sampled frame pairs and threshold it by Otsu’s
method [4] into a dynamic region and its complement.
The static region is then reduced by a fixed morpholog-
ical erosion and a 4% frame border discarded; together
these form the ignored transition band, so boundary
pixels—where the partition is least certain and flow
least reliable—contribute to neither region. The same
procedure applies to both tracks.

Why the estimate is restricted to an early window.
Deriving a partition from a sequence’s entire motion
field would be circular, since accumulated background
drift would itself be read as evidence that the affected
region is dynamic. Restricting the estimate to the early
window breaks that circularity in the direction that
matters: at the 60s horizon the partition is fixed from
the first 7.1s, before long-horizon drift has accumu-
lated, and the metrics are evaluated over the remain-
der.

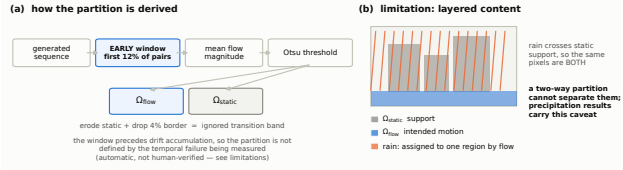


Figure 1. Region-partition protocol. (a) The partition is derived automatically from early-window flow, with an ignored transition band at the boundary. (b) A two-way partition cannot represent layered content such as rain crossing static support, which is the scope limit stated in the main paper.

Limitations. The partition is obtained automatically rather than by annotation, so no inter-annotator agreement is reported, and for the image-conditioned track it is derived per generated sequence rather than once from the shared source image. The early window is a weaker guarantee than a first-frame annotation: a system that drifts within the first 12% of its rollout can influence its own partition. We therefore treat partitions as measured quantities and report the consequence directly—every measurement is recomputed under systematic erosion and dilation of the boundary, with the response in Table 1.

3. Validation Protocol

Geometric severity is specified in a common physical unit—the mean pixel displacement a corruption induces over the frame by the end of the rollout—and each family’s parameter is derived from it using the frame geometry: a translation of d pixels, a rotation of $d/\bar{r}$ radians, or a zoom of $1 + d/\bar{r}$, where $\bar{r}$ is the mean distance from the frame centre. This is what makes the geometric families comparable. A parameter in degrees is not commensurate with one in pixels: at the evaluation resolution one degree of rotation induces roughly 2.7 px of mean displacement, so families compared by their native parameters differ in injected severity by more than an order of magnitude, and a factor can appear non-monotone simply because the corruption was too weak to clear the clip’s own baseline drift.

4. Partition and Composition Robustness

Two questions decide whether the region partition can be trusted as a measurement basis: whether a system can be advantaged by receiving an easier mask, and whether the ordering depends on the one scene category the partition cannot represent.

The static region occupies a similar share of the frame for every audited system, between 0.61 and 0.66 on average, so no system is scored over a markedly

Factor	target	ρ	ρ_{rot}	off-target	admitted
fBD	drift	+1.00	+0.70	14%	✓
NBF	drift	+1.00	+0.60	6%	✓
MCFF-L	freeze	+1.00	n/a	14%	✓
FP	freeze	+0.80	n/a	21%	✓
DLR	drift	+1.00	-0.80	5%	✓
DAR	drift	+0.90	+1.00	18%	✓
VBench DD	drift	+0.70	+0.70	2%	—

Table 1. Mechanistic validation and factor admission. Each factor is scored on the corruption its definition says it should detect: accumulating global drift for the static-fidelity and drift diagnostics, progressive freezing for the persistence factors. ρ is Spearman correlation against injected severity, signed so that a correct response is positive; ρ_{rot} repeats it for rotational drift, the harder case. off-target is the largest relative excursion under corruptions the factor should ignore—photometric drift and perturbation of the region partition—so a small value indicates selectivity. Admission requires $\rho \geq 0.7$ and off-target response below 25% on the factor’s target family. Rotational response is reported but not gated, and the factors divide that work explicitly: DLR is a translational-leakage diagnostic whose rotational response is inverted (−0.80), while rotational drift is carried by fBD and DAR at +0.70 and +1.00. That division is why DLR is never read alone. VBench DD is a comparison row rather than a candidate factor—it is put through the identical corruptions to show what a whole-frame score does under them—so its admission cell is left blank. Twelve reference clips per severity level.

larger or smaller support than another. Across systems, that area correlates negatively with feature-based drift. Over seven systems the correlation is not determined, but its sign is the one circularity would produce—a system that drifts within the estimation window enlarges its own dynamic region and shrinks the support its drift is measured over—and we report it as an open validity question rather than dismissing it on significance. The decisive test, recomputing every score under a region partition not derived from the same sequence, is stated as required future work in Sec. 2; we do not claim to have run it.

Removing precipitation, where rain and snow cross static support and a two-way partition must assign those pixels to one region by flow magnitude alone, leaves the normalised-background-flow ordering unchanged and moves only mid-field positions under feature-based drift.

5. Paired Differences

Separation claims in the main text are made on the bootstrap distribution of per-prompt differences, not on whether two marginal intervals overlap. Every system sees the same prompts, so the paired difference is the statistic the design supports: overlapping marginal

System	static-region area
CausVid	0.663 ± 0.073
Reward-Forcing	0.661 ± 0.060
Self-Forcing	0.651 ± 0.087
Infinite-Forcing	0.650 ± 0.090
LongLive	0.644 ± 0.095
Rolling-Forcing	0.616 ± 0.107
Causal-Forcing	0.614 ± 0.086
Does mask area predict the drift score?	
Spearman(area, fBD) over 7 systems	-0.75, exact $p = 0.066$
Ordering with precipitation excluded (3 of 23 prompts)	
fBD ordering	best and worst unchanged
NBF ordering	identical

Table 2. Partition and composition robustness. The static region occupies a similar share of the frame for every audited system, so no system is scored over a much larger or smaller support than another. The area does correlate negatively with drift across systems; over seven systems that correlation is not determined, but its sign is the one circularity would produce—a system that drifts early enlarges its own dynamic region—and we flag it as an open validity question rather than dismissing it. Removing precipitation, the category a two-way partition cannot represent, leaves the normalised-background-flow ordering unchanged.

intervals can conceal a reliable paired difference, and non-overlapping ones can suggest a difference that is not there.

6. Generation Configuration

Every audited checkpoint was run through a single common inference configuration, recorded in full and released with the benchmark: a four-step denoising schedule (1000/750/500/250), timestep shift 5.0, guidance scale 5.0, six frames per block, and a fixed seed. Only the checkpoint path and the target horizon differ between runs.

We state plainly what this does and does not license. It is a matched-setting comparison, and it is reproducible precisely because the setting is uniform and recorded. It is not an audit of each method under its authors’ released defaults, and we no longer describe it as one. The cost is real: several of these checkpoints are distilled for a specific step budget, and evaluating them at a common four-step schedule does not observe them at their own operating point. Results are therefore statements about these checkpoints under this configuration. Re-running the sweep at each method’s released defaults, and reporting how much the ordering moves, is the natural next version of this audit.

The image-conditioned roster is recorded the same way. Rows marked [†] did not reach these horizons under their released configuration and were run instead through a common long-horizon wrapper, identical for every such row: six frames per block, image-conditioned, with the released checkpoint otherwise un-

modified. That wrapper is a property of this evaluation, not of any audited system, which is why its rows are read as deployment sensitivity and are never ranked against native rows. Its per-method sampler settings were not captured either, and the same limitation therefore applies.

7. Measurement Coverage

Coverage is counted from valid per-video measurements rather than from the presence of a results file, so a run that terminated early is reported as partial. Separately, fBD is feature-based [6] and declines to report where too few repeatable keypoints survive; those abstentions are tabulated rather than folded silently into a smaller n .

8. Complete Audit, Both Tracks

The main paper reports the text-conditioned track at 60s, the tier whose sample supports method-level separation. Both tracks appear here at every horizon, with uncertainty from a percentile bootstrap over items [1]. For the image-conditioned track, systems marked matched run under a common long-horizon wrapper rather than their released configuration, so their scores are read as deployment sensitivity rather than as the published method’s performance.

9. Aggregation Sensitivity

The evaluation set is dominated by one flow medium, so the audit must be shown not to be a report on that

Comparison	n	mean diff.	95% CI
Infinite-Forcing – Causal-Forcing	23	-83.98	$[-102.59, -65.52]$
Infinite-Forcing – LongLive	23	-10.22	$[-14.72, -6.18]$
Infinite-Forcing – Self-Forcing	23	-9.07	$[-13.29, -5.53]$
Infinite-Forcing – Rolling-Forcing	23	-7.58	$[-10.38, -5.12]$
Infinite-Forcing – CausVid	23	-5.62	$[-9.61, -2.10]$
Infinite-Forcing – Reward-Forcing	23	-4.86	$[-7.48, -2.69]$

Table 3. Paired differences in NBF, 60s horizon. Every system is evaluated on the same prompts, so separation is judged on the bootstrap distribution of per-prompt differences (10k resamples, fixed seed) rather than on whether two marginal intervals happen to overlap—overlapping intervals can hide a reliable paired difference, and non-overlapping ones can suggest a difference that is not. 6 of 6 intervals exclude zero.

System	setting	resolution	fps	horizons
CausVid	native	832×480	16	5s, 60s, 120s, 240s
Self-Forcing	native	832×480	16	5s, 60s, 120s, 240s
Infinite-Forcing	native	832×480	16	5s, 60s, 120s, 240s
Rolling-Forcing	native	832×480	16	5s, 60s, 120s, 240s
Reward-Forcing	native	832×480	16	5s, 60s, 120s, 240s
LongLive	native	832×480	16	5s, 60s, 120s, 240s
Causal-Forcing	native	832×480	16	5s, 60s, 120s, 240s

Table 4. Recorded generation configuration (T2V). Every audited system was run from its released checkpoint. Native output formats are not identical across the roster—resolutions and frame rates differ where the releases differ—and the comparison is unaffected because the factors normalise them rather than assume them away: NBF is expressed per second and per frame width, fBD as a percentage of the frame diagonal, and every sequence is resampled to a common flow-estimation rate before measurement. Per-method sampler settings—denoising steps, guidance scale, block length and seed—are uniform across the roster and recorded with the benchmark: a four-step schedule, guidance 5.0, six frames per block, fixed seed. This is a matched-setting comparison, reproducible because the setting is uniform, and it does not observe a checkpoint distilled for a different step budget at its own operating point.

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medium alone. The main paper aggregates by averaging over prompts. The alternative, weighting each scene category equally, is not preferable here: at the principal horizon three of the five populated categories contain two prompts each, so a macro-average would give a two-prompt cell the same influence as a fourteen-prompt one and inflate variance in exchange for nominal balance. We therefore report both and state what changes.

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The best- and worst-ranked systems are the same under either aggregation, so the contrast the paper draws does not depend on the choice. Intermediate ranks do move, which is why no ordinal claim is made about the middle of the field.

10. Whole-Frame and SNF-Bench Rank Correlation 200 201

11. Deployment Sensitivity 202

12. Signed Drift Attenuation 203

Drift attenuation is stored signed and reported clipped to $[0, 1]$. Negative values are informative rather than anomalous: where local flow opposes the estimated global field, compensation increases the measured magnitude. We report their incidence rather than suppressing it. 204
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13. Evaluation-Set Composition 210

14. Exploratory Perceptual Pilot 211

This section supports no perceptual-validity claim. It reports an exploratory pilot that did not establish agreement between human judgments and the factor orderings, and it is included because the reason it failed is informative and because the protocol is what a powered study should start from. Nothing here ranks sys- 212
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System	setting	resolution	fps	horizons
Causal-Forcing++ (2-step, frame-wise)	native	832×480	16	5s, 60s, 120s, 240s
Wan2.1-I2V-14B-480P	native	832×464	16	5s
Wan2.2-I2V-A14B	native	832×464	16	5s
LTX-Video 13B-0.9.8-distilled	native	832×480	24	5s
Causal-Forcing++ (1-step, frame-wise)	native	832×480	16	60s, 120s
Causal-Forcing++ (2-step) [†]	matched	832×480	16	5s, 60s, 120s, 240s
Causal-Forcing++ (1-step) [†]	matched	832×480	16	5s, 60s, 120s, 240s
Causal-Forcing (frame-wise) [†]	matched	832×480	16	5s, 60s, 120s, 240s
Self-Forcing [†]	matched	832×480	16	5s, 60s, 120s, 240s
CausVid [†]	matched	832×480	16	5s, 60s, 120s, 240s

Table 5. Recorded generation configuration (I2V). Every audited system was run from its released checkpoint. Native output formats are not identical across the roster—resolutions and frame rates differ where the releases differ—and the comparison is unaffected because the factors normalise them rather than assume them away: NBF is expressed per second and per frame width, fBD as a percentage of the frame diagonal, and every sequence is resampled to a common flow-estimation rate before measurement. Per-method sampler settings—denoising steps, guidance scale, block length and seed—are uniform across the roster and recorded with the benchmark: a four-step schedule, guidance 5.0, six frames per block, fixed seed. This is a matched-setting comparison, reproducible because the setting is uniform, and it does not observe a checkpoint distilled for a different step budget at its own operating point. Rows marked [†] ran under the common long-horizon wrapper rather than a released configuration and are grouped separately below the native rows.

track	horizon	measured	fBD abstained	rate
I2V	120s	400	0	0.0%
I2V	240s	65	0	0.0%
I2V	5s	180	0	0.0%
I2V	60s	630	7	1.1%
T2V	120s	42	0	0.0%
T2V	240s	28	0	0.0%
T2V	5s	48	0	0.0%
T2V	60s	185	0	0.0%

Table 6. Measurement coverage and fBD abstention. fBD is feature-based and declines to report when too few repeatable keypoints survive in the static region. Every abstention in this benchmark occurs on a single desert dust-storm scene, whose flat, dust-obscured ground offers no stable structure to track, and it occurs there for every system alike—so it reflects the scene, not a method. The remaining factors are reported for those clips as usual; only fBD is withheld.

tems, and no result from it enters any factor, table or figure elsewhere in this work.

The pilot ran 24 pairs per rater—twelve on each axis—with eight raters, of whom four were independent of the authors and four were the authors’ own sessions, reported separately. After majority voting across raters, three pairs on the static axis and five on the persistence axis remained decided. Those counts, not the percentages they produce, are the honest summary of this run.

The question the pilot asked is whether independent

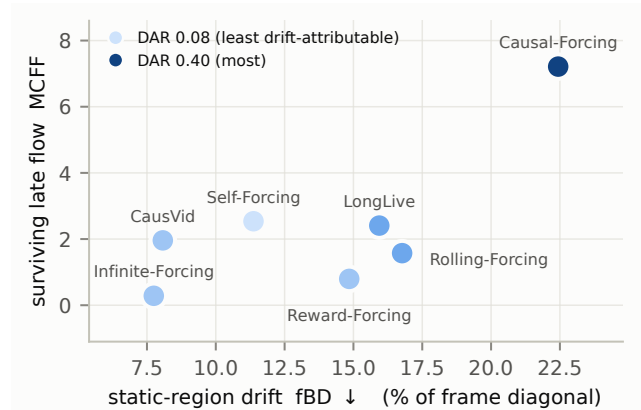


Figure 2. Operating regimes under native deployment. Each system in the static-drift-surviving-flow plane. Colour encodes the relative attenuation of measured dynamic-region flow after global-motion compensation, not a share of it attributable to drift: the quantity is signed, and values are shown clipped to $[0, 1]$. The view exposes trade-offs; it is not a leaderboard.

judges order clip pairs the way the factors do. That is metric validation rather than preference, so a small sample would have been survivable had the pairs been constructed correctly; as Sec. 14 reports below, they were not.

Ethics. Raters were volunteers drawn from the authors’ institution and immediate colleagues, and four of the eight sessions were the authors’ own pilot runs, re-

Method	status	setting	5s	60s	120s	240s
CausVid	public	native	V6 T6 B	V23 T23 B	V6 T6 B	V4 T4 B
Self-Forcing	public	native	V6 T6 B	V23 T23 B	V6 T6 B	V4 T4 B
Infinite-Forcing	public	native	V6 T6 B	V23 T23 B	V6 T6 B	V4 T4 B
Rolling-Forcing	public	native	V6 T6 B	V24 T24 B	V6 T6 B	V4 T4 B
Reward-Forcing	public	native	V12 T12 B	V23 T23 B	V6 T6 B	V4 T4 B
LongLive	public	native	V6 T6 B	V23 T23 B	V6 T6 B	V4 T4 B
Causal-Forcing	public	native	V6 T6 B	V23 T23 B	V6 T6 B	V4 T4 B

Table 7. Asset and score coverage for the audited public systems. Tn/N marks entries where a metrics file exists but only n of N videos hold usable measurements.

Method	fBD↓	NBF↓	MCFF-E	MCFF-L↑	FP↑	DLR↓	DAR↓
5s horizon ($n = 6$)							
CausVid	2.08	6.56	1.44	2.91	1.24	0.26	0.07
Self-Forcing	0.68	10.8	7.70	4.35	0.76	0.49	0.03
Infinite-Forcing	0.24	2.32	1.23	0.89	1.08	0.21	0.09
Rolling-Forcing	3.51	12.1	3.10	3.69	1.32	0.48	0.17
Reward-Forcing	1.10	5.21	0.63	0.58	1.37	0.44	0.28
LongLive	1.91	7.31	1.25	2.27	1.16	0.59	0.32
Causal-Forcing	9.55	31.0	5.32	3.01	1.13	0.62	0.37
60s horizon ($n = 23$)							
CausVid	8.08	9.27	2.87	1.96	0.67	0.45	0.14
Self-Forcing	11.4	12.7	10.4	2.54	0.44	0.50	0.14
Infinite-Forcing	7.75	3.65	1.17	0.29	0.63	0.55	0.14
Rolling-Forcing	16.8	11.2	3.18	1.57	0.75	0.54	0.18
Reward-Forcing	14.8	8.51	3.81	0.80	0.49	0.60	0.17
LongLive	15.9	13.9	6.46	2.41	0.75	0.59	0.17
Causal-Forcing	22.4	87.6	9.84	7.21	0.83	0.94	0.43
120s horizon ($n = 6$)							
CausVid	6.15	3.42	0.36	0.52	1.06	0.41	0.13
Self-Forcing	21.2	6.80	4.85	0.82	0.85	0.75	0.15
Infinite-Forcing	16.1	2.60	0.50	0.32	0.64	0.57	0.21
Rolling-Forcing	23.8	10.6	3.33	0.77	0.38	0.82	0.22
Reward-Forcing	20.0	11.6	2.05	0.69	0.39	0.70	0.12
LongLive	22.5	9.66	5.10	1.19	0.81	0.84	0.23
Causal-Forcing	21.8	130	10.5	10.7	0.90	0.81	0.50
240s horizon ($n = 4$)							
CausVid	7.23	13.4	0.42	0.87	0.99	0.85	0.08
Self-Forcing	20.0	17.4	3.09	0.50	0.15	2.26	0.20
Infinite-Forcing	20.0	4.15	0.33	0.23	1.02	1.08	0.15
Rolling-Forcing	23.7	15.4	2.15	0.97	0.46	0.78	0.18
Reward-Forcing	26.4	11.4	1.09	3.55	1.53	0.50	0.07
LongLive	23.1	11.8	1.52	1.32	1.20	1.38	0.31
Causal-Forcing	18.9	102	4.72	6.90	1.21	0.73	0.58

Table 8. Complete T2V audit, all four horizons. The main paper reports the 60s and 120s panels; the 5s tier is an initialisation check and 240s a diagnostic extreme. Every row is computed under the same frozen specification, so panels are comparable within a horizon; prompt sets differ across horizons, so columns are not comparable between panels. All systems are evaluated in the native setting. n is stated per horizon and is common to every row of that panel.

ported separately and excluded from the primary analysis. Participation was voluntary and uncompensated, no personal data beyond a self-chosen identifier was collected, and no clip depicts an identifiable person. The study involved no intervention and no sensitive

material. Because a powered study will recruit outside this group, we will obtain the institutional review or documented exemption that such recruitment requires before running it.

Method	setting	fBD↓	NBF↓	MCFF-E	MCFF-L↑	FP↑	DLR↓	DAR↓
5s horizon ($n = 10$)								
Causal-Forcing++ (2-step)	matched	3.69	10.0	1.06	0.61	0.78	0.67	0.32
Causal-Forcing++ (1-step)	matched	5.28	11.4	1.05	0.47	0.64	0.66	0.29
Causal-Forcing (frame-wise)	matched	16.1	45.3	8.48	2.79	0.43	0.68	0.29
Causal-Forcing++ (2-step, frame-wise)	native	1.22	10.6	2.23	0.78	0.47	0.68	0.20
Self-Forcing	matched	9.66	5.74	0.69	0.26	0.70	0.74	0.21
CausVid	matched	15.2	10.8	0.76	0.87	0.98	1.04	0.10
Wan2.1-I2V-14B-480P	native	5.01	21.3	6.67	5.58	1.09	0.52	0.18
Wan2.2-I2V-A14B	native	6.31	25.3	4.33	9.09	0.95	0.50	0.27
LTX-Video 13B-0.9.8-distilled	native	6.65	17.8	3.59	3.48	0.92	0.39	0.24
60s horizon ($n = 30$)								
Causal-Forcing++ (2-step)	matched	9.18	5.97	0.85	0.60	0.86	0.67	0.25
Causal-Forcing++ (1-step)	matched	5.34	7.05	0.86	0.57	0.77	0.64	0.28
Causal-Forcing (frame-wise)	matched	21.1	36.4	2.80	2.69	0.89	0.80	0.35
Causal-Forcing++ (2-step, frame-wise)	native	9.94	6.91	1.30	0.78	1.02	0.75	0.17
Self-Forcing	matched	17.3	5.52	1.50	0.69	0.89	0.67	0.18
CausVid	matched	19.9	3.31	0.58	0.41	0.81	0.57	0.14
Causal-Forcing++ (1-step, frame-wise)	native	10.4	5.25	0.53	0.52	1.01	0.83	0.22
120s horizon ($n = 20$)								
Causal-Forcing++ (2-step)	matched	9.65	8.81	0.89	0.82	0.93	0.58	0.28
Causal-Forcing++ (1-step)	matched	7.26	9.39	0.84	0.66	0.86	0.62	0.31
Causal-Forcing (frame-wise)	matched	19.9	52.7	4.82	5.06	1.08	0.67	0.24
Causal-Forcing++ (2-step, frame-wise)	native	11.4	17.3	1.89	1.54	1.01	0.75	0.09
Self-Forcing	matched	23.0	10.3	2.41	0.76	0.45	0.66	0.26
CausVid	matched	19.7	6.04	1.15	0.83	0.67	0.48	0.14
Causal-Forcing++ (1-step, frame-wise)	native	13.6	7.39	1.01	0.35	0.64	0.77	0.33
240s horizon ($n = 5$)								
Causal-Forcing++ (2-step)	matched	10.4	4.83	0.53	0.59	1.05	0.55	0.15
Causal-Forcing++ (1-step)	matched	6.42	6.66	0.56	0.51	0.83	0.67	0.30
Causal-Forcing (frame-wise)	matched	19.8	31.5	3.08	2.28	0.74	0.69	0.45
Causal-Forcing++ (2-step, frame-wise)	native	6.32	3.95	1.23	0.68	0.67	0.83	0.02
Self-Forcing	matched	22.3	5.74	1.48	1.59	0.81	0.71	0.08
CausVid	matched	18.4	2.08	0.81	0.79	0.86	0.44	0.11

Table 9. Complete I2V audit, all four horizons. The main paper reports the 60s and 120s panels; the 5s tier is an initialisation check and 240s a diagnostic extreme. Every row is computed under the same frozen specification, so panels are comparable within a horizon; prompt sets differ across horizons, so columns are not comparable between panels. n is stated per horizon and is common to every row of that panel.

246 Questions. One forced choice per axis, with an ex-
 247 plicit third option: static stability—in which clip does
 248 the background stay more fixed in place—and motion
 249 persistence—in which clip does the intended motion
 250 keep going rather than fade or freeze. Judges are never
 251 asked which output is better, or which looks more nat-
 252 ural. “Better” conflates the two failures the benchmark
 253 separates, and naturalness is outside the stated scope,
 254 so either question would produce a result the paper
 255 could not own.

256 Pairs. A pair is two systems’ outputs on the same
 257 prompt at the same horizon, never across prompts.
 258 Pairs are stratified by the relative gap in the axis’s own

factor: clear-gap pairs, where the factor separates the
 two outputs, carry the agreement claim, while near-tie
 pairs test the opposite property—that where the fac-
 tors do not separate, judges also cannot tell the clips
 apart. Presentation side is randomised per trial and re-
 solved back to a system identity only after the response
 is recorded; system names are never shown.

Statistics. The unit of analysis is the pair, majority-
 voted across raters, and intervals are percentile boot-
 strap resampling pairs. Responses are clustered inside
 rater and inside pair, so an interval computed over re-
 sponses would claim precision the design does not have.
 We report inter-rater agreement so the reader can see

System	prompt mean	rank	macro mean	rank
NBF				
Infinite-Forcing	3.65	1	4.12	1
Reward-Forcing	8.51	2	10.11	4 [†]
CausVid	9.27	3	9.93	3
Rolling-Forcing	11.23	4	11.32	5 [†]
Self-Forcing	12.72	5	8.98	2 [†]
LongLive	13.86	6	14.69	6
Causal-Forcing	87.62	7	61.70	7
fBD				
Infinite-Forcing	7.75	1	6.10	1
CausVid	8.08	2	9.51	3 [†]
Self-Forcing	11.37	3	9.27	2 [†]
Reward-Forcing	14.85	4	15.61	5 [†]
LongLive	15.94	5	14.93	4 [†]
Rolling-Forcing	16.77	6	18.85	6
Causal-Forcing	22.44	7	23.49	7

Table 10. Sensitivity of the audit to category aggregation. Prompt means, as reported in the main paper, against category macro-averages, for the two static-fidelity factors at the principal horizon. [†] marks a system whose rank changes. The best- and worst-ranked systems are identical under both, so the paper’s headline contrast does not depend on the choice; intermediate positions do change, which is why no ordinal claim is made about the middle of the field.

generic metric	fBD	NBF	MCFF	DLR
VB-DD	+0.75	+0.93	+0.89	+0.36
VB-bg	-0.64	-0.71	-0.75	-0.50
VB-smooth / VB-flick	-0.96	-0.82	-0.61	-0.64

Table 11. Rank disagreement at 60s. Spearman correlation over $n = 7$ method means—descriptive, not inferential. Positive correlation between apparent motion and background flow is the effect the benchmark isolates. Smoothness and flickering rank these systems identically and share a row.

($\alpha = 0.18$ and 0.24 on the two axes), and the direction of the estimate reverses depending on which rater subset is used. We therefore report the counts and the interval rather than a headline percentage, and we do not claim that perceptual agreement has been established.

The cause is a property of the pair set rather than of the raters, and it is measurable. The pairs were stratified by the gap in each axis’s own factor but were not isolated against the other axis. On the static question the median relative gap in the target factor is 0.29 while the median gap in the persistence factor is 0.58 : the clips shown differ about twice as much on the axis the rater was told to ignore as on the one being asked about, and on half the pairs both axes differ substantially. A rater seeing one clip drift and the other stall cannot report which axis drove the answer, which is exactly the condition that produces low inter-rater agreement and near-chance ordering. A powered study therefore requires axis-isolated pairs—large gap in the target factor, small gap in the other—and that constraint, not rater count, is the first thing to fix.

Outcome of the vision–language arm: negative, with a located cause. The same pairs and the same wording were put to an open-weight judge [5] at temperature zero, each pair asked in both presentation orders. Two controls precede any reading of its answers. Shown one clip against itself, where the only correct answer is that they are the same, it answered correctly on six of six.

Method	Generic evidence	SNF-Bench evidence	Interpretation
CausVid	VB-DD rank 6/7	NBF rank 3/7, MCFF-L rank 4/7	stable sequence with decaying intended motion
Causal-Forcing	VB-DD rank 1/7	NBF rank 7/7, MCFF-L rank 1/7	substantial apparent motion, strongly contaminated by background displacement
Infinite-Forcing	VB-DD rank 7/7	NBF rank 1/7, MCFF-L rank 7/7	stable sequence with decaying intended motion
LongLive	VB-DD rank 3/7	NBF rank 6/7, MCFF-L rank 3/7	substantial apparent motion, strongly contaminated by background displacement
Reward-Forcing	VB-DD rank 5/7	NBF rank 2/7, MCFF-L rank 6/7	stable sequence with decaying intended motion
Self-Forcing	VB-DD rank 2/7	NBF rank 5/7, MCFF-L rank 2/7	substantial apparent motion, strongly contaminated by background displacement

Table 12. Interpretation changes between generic whole-frame metrics and SNF-Bench at 60s. Neither evaluation is labelled correct; the table identifies information hidden by whole-frame aggregation. Cases are selected by a fixed rank-gap rule applied to all public methods, so none can be selectively omitted.

Checkpoint	horizon	Δ fBD	Δ NBF	Δ MCFF-L	Δ DLR
Causal-Forcing++ (2-step)	5s	+2.47	-0.61	-0.17	-0.01
Causal-Forcing++ (2-step)	60s	-0.76	-0.95	-0.17	-0.08
Causal-Forcing++ (2-step)	120s	-1.71	-8.46	-0.73	-0.17
Causal-Forcing++ (2-step)	240s	+4.06	+0.88	-0.09	-0.28
Causal-Forcing++ (1-step)	60s	-5.05	+1.80	+0.05	-0.20
Causal-Forcing++ (1-step)	120s	-6.34	+2.00	+0.31	-0.14

Table 13. Deployment sensitivity. Change from each checkpoint’s own released setting to the common long-horizon wrapper, for the two checkpoints evaluated both ways. Absolute matched scores are never read as the published method’s performance.

Method	track	dur	n	negative	rate	most negative
CausVid	i2v	120s	20	1	5.0%	-0.055
CausVid	i2v	5s	10	5	50.0%	-1.076
CausVid	i2v	60s	30	2	6.7%	-0.084
Causal-Forcing++ (1-step)	i2v	120s	20	1	5.0%	-0.027
Causal-Forcing (frame-wise)	i2v	120s	20	3	15.0%	-0.171
Causal-Forcing++ (2-step)	i2v	5s	10	1	10.0%	-0.164
Causal-Forcing++ (1-step, frame-wise)	i2v	60s	30	3	10.0%	-1.420
Causal-Forcing++ (2-step, frame-wise)	i2v	120s	20	5	25.0%	-0.241
Causal-Forcing++ (2-step, frame-wise)	i2v	240s	5	1	20.0%	-0.651
Causal-Forcing++ (2-step, frame-wise)	i2v	5s	10	1	10.0%	-0.288
Causal-Forcing++ (2-step, frame-wise)	i2v	60s	30	1	3.3%	-0.001
LTX-Video 13B-0.9.8-distilled	i2v	5s	10	2	20.0%	-0.018
Self-Forcing	i2v	120s	20	2	10.0%	-0.046
Self-Forcing	i2v	240s	5	1	20.0%	-0.136
Self-Forcing	i2v	60s	30	4	13.3%	-0.578
Wan2.1-I2V-14B-480P	i2v	5s	10	1	10.0%	-0.255
Wan2.2-I2V-A14B	i2v	5s	10	1	10.0%	-0.335
CausVid	t2v	240s	4	1	25.0%	-0.384
CausVid	t2v	5s	6	2	33.3%	-0.093
Causal-Forcing	t2v	5s	6	2	33.3%	-0.273
Causal-Forcing	t2v	60s	23	3	13.0%	-0.640
Infinite-Forcing	t2v	240s	4	1	25.0%	-0.158
Infinite-Forcing	t2v	60s	23	4	17.4%	-0.359
LongLive	t2v	120s	6	1	16.7%	-1.076
LongLive	t2v	5s	6	1	16.7%	-0.139
LongLive	t2v	60s	23	2	8.7%	-0.051
Reward-Forcing	t2v	120s	6	1	16.7%	-0.116
Reward-Forcing	t2v	5s	12	2	16.7%	-0.145
Reward-Forcing	t2v	60s	23	3	13.0%	-0.317
Rolling-Forcing	t2v	120s	6	3	50.0%	-0.353
Rolling-Forcing	t2v	5s	6	1	16.7%	-0.068
Rolling-Forcing	t2v	60s	24	2	8.3%	-0.038
Self-Forcing	t2v	120s	6	2	33.3%	-0.130
Self-Forcing	t2v	240s	4	1	25.0%	-0.153
Self-Forcing	t2v	5s	6	1	16.7%	-0.046
Self-Forcing	t2v	60s	23	4	17.4%	-1.117

Table 14. Incidence of negative DAR, i.e. clips where global-motion compensation increased measured dynamic-region flow. 21 further entries had no negative values and are omitted.

318	Shown a frozen single frame against a real sequence—	slots were chosen 8 and 12 times. The judge is therefore	322
319	the largest difference available on the motion axis—its	well behaved and still cannot do the task: it answered	323
320	verdict followed the swapped clip on only two of six.	“indistinguishable” on most pairs the factors separate	324
321	Presentation bias is absent: over 48 queries the two	clearly, leaving one and two decided pairs on the two	325

track/duration	n	river_stream	ocean_waves	precipitation	fire_smoke	lava_volcanic	windborne	cats
i2v/5s	10	2	.	3	2	1	2	5
i2v/60s	30	9	4	9	1	3	4	6
i2v/120s	20	6	3	3	3	1	4	6
i2v/240s	5	1	2	1	.	1	.	4
t2v/5s	6	3	1	1	1	.	.	4
t2v/60s	23	14	2	3	2	.	2	5
t2v/120s	6	4	.	1	.	.	1	3
t2v/240s	4	2	.	1	.	.	1	3

Table 15. Scene-category balance of the evaluation set, by track and horizon. A dot marks a category absent from that cell.

Axis	Judge	agrees with factor	called indistinguishable	pairs	inter-rater α
Static stability (fBD)	human	– (3 decided)	–	8	0.18
	vision–language	– (1 decided)	1/1	7	–
Motion persistence (MCFF-L)	human	80% [40, 100]	–	8	0.24
	vision–language	– (2 decided)	–	3	–

Table 16. Preliminary perceptual validation of the factor axes. Each judge answers one question per axis and never which output is better. agree is agreement with the factor’s ordering on pairs the factor separates clearly, over pairs the judge decided; can’t tell is the share of near-tie pairs called indistinguishable, which is the correct answer there. The unit is the pair, majority-voted across raters, and the interval is a percentile bootstrap resampling pairs rather than responses, which are clustered within rater and pair. A dash marks a cell with too few decided pairs to carry a rate; the count is given instead. This is a preliminary check on a measurement instrument, not a powered study, and no system is ranked by it.

326 axes, which is far too few to support an agreement rate.
 327 We report none.

328 The controls locate the cause in the judge rather
 329 than in the factors or the prompt. A model that can-
 330 not separate a still image from moving water is not
 331 positioned to separate two rollouts that differ subtly.
 332 This says that an open-weight judge at this scale can-
 333 not presently substitute for human raters on these axes.
 334 It does not say that stronger judges would fail, and we
 335 make no claim about them: contemporary generation
 336 benchmarks that report successful automated judging
 337 use frontier proprietary models [2, 7] or scorers fine-
 338 tuned for the task [3], and we have no evidence either
 339 way about those here.

340 Status. Both arms are reported as preliminary and
 341 neither supports an agreement claim. What the pilot
 342 does establish is that the instrument runs end to end,
 343 that raters engage with the axis questions rather than
 344 defaulting to a side, and that the binding constraint
 345 on a powered study is pair construction rather than
 346 rater count: axis-isolated pairs, a near-tie stratum, and
 347 raters recruited outside the authors. Protocol, pair
 348 manifest and judge prompt are released verbatim so
 349 the check can be repeated and enlarged. To restate
 350 the scope without qualification: what is released here
 351 is a validated protocol and a diagnosis of why this run
 352 of it failed. No conclusion about perceptual agreement,
 353 in either direction, should be drawn from the counts in
 354 this section. They are reported so the pilot is auditable,

not because they support an inference.

15. Additional Qualitative Material

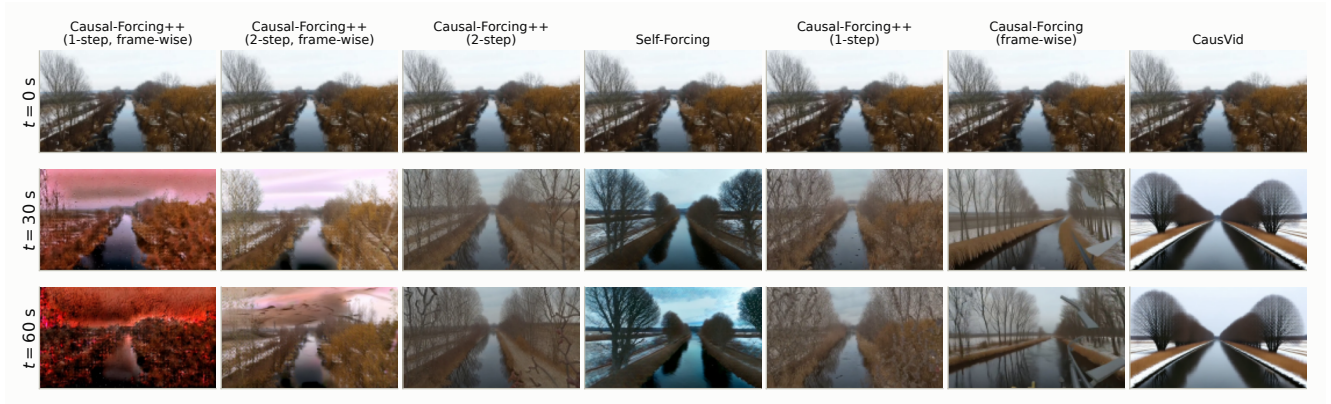


Figure 3. Image-conditioned systems on one shared source image. Every system begins from the identical frame, so divergence over the rollout is attributable to the system rather than to a differently imagined scene—which is not true of the text-conditioned track, where each system authors its own layout. Columns are ordered by increasing static-region drift.

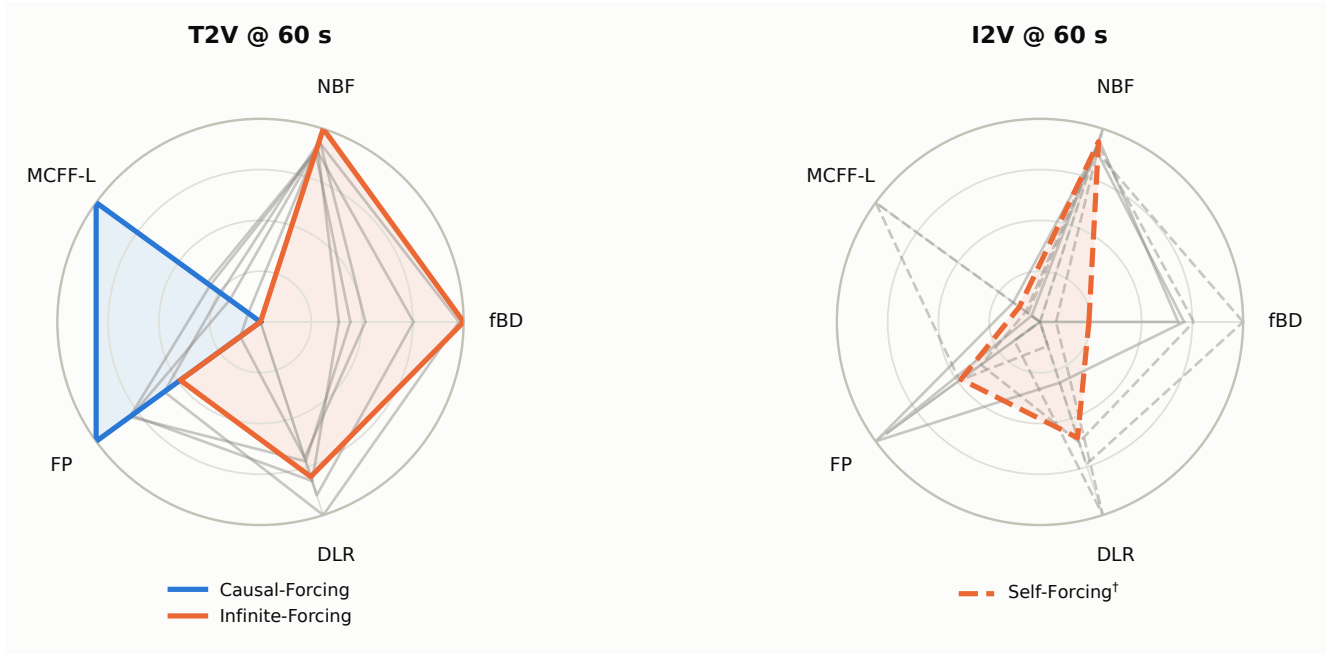


Figure 4. Factor profile per track. Each factor is rescaled across the audited systems and oriented so that outward is always better; plotting raw values would place “most drift” and “most surviving motion” both outward and the shape would carry no meaning. No system encloses the others on either track, which is the conclusion the operating-regime view reaches, shown as a profile. On the image-conditioned panel, dashed outlines marked [†] are rows run under the common long-horizon wrapper rather than a released configuration: they are shown for shape, and are not peers of the solid native outlines. The rescaling is relative to the audited roster, so a profile is descriptive of this set and would change if the set changed.

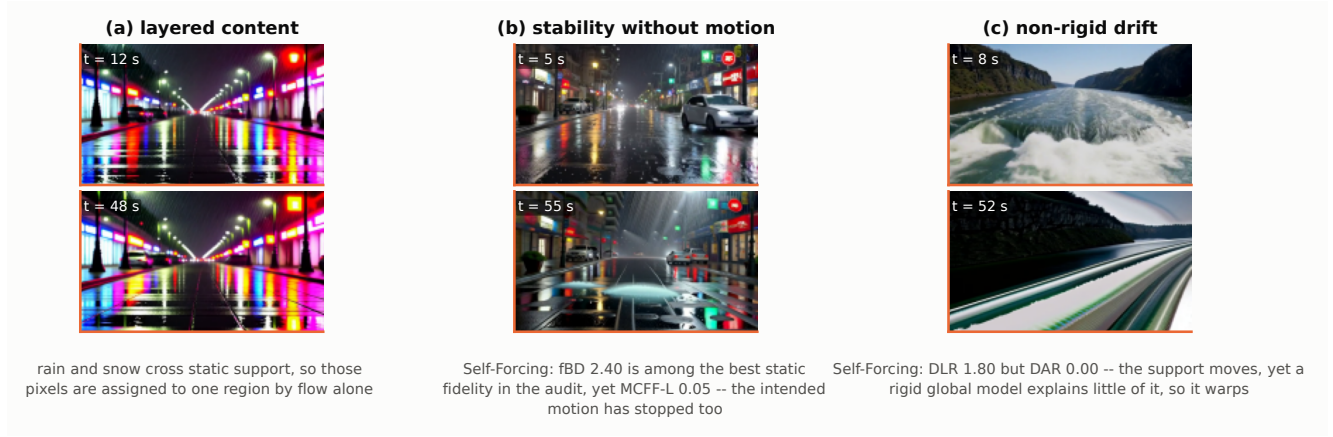


Figure 5. Stated scope boundaries. (a) A two-way partition cannot separate precipitation from the support it crosses. (b) A system can attain near-best static fidelity precisely because its intended motion has also stopped. (c) High leakage with low attenuation marks static-region motion that a rigid global similarity model does not account for; what the residual consists of is not identified by these measurements.



Figure 6. Every audited text-conditioned system on one prompt. Columns are ordered by increasing static-region drift; the two highlighted systems are the pair a whole-frame motion score cannot separate.

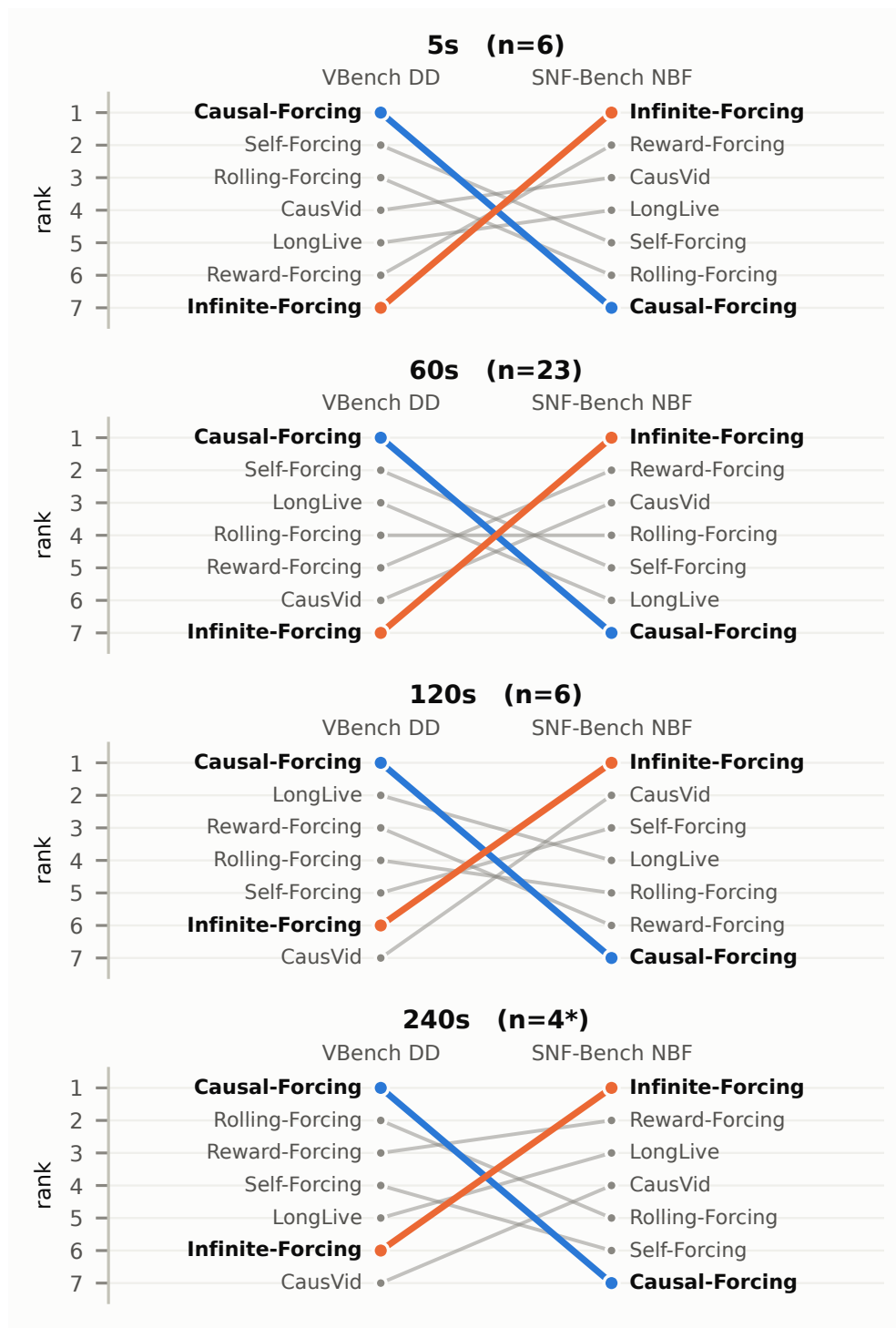


Figure 7. Rank disagreement at every audited horizon. Rank under a whole-frame motion score against rank under normalised background flow. Both quantities are independent of the global-motion estimator; prompt counts are given per panel.

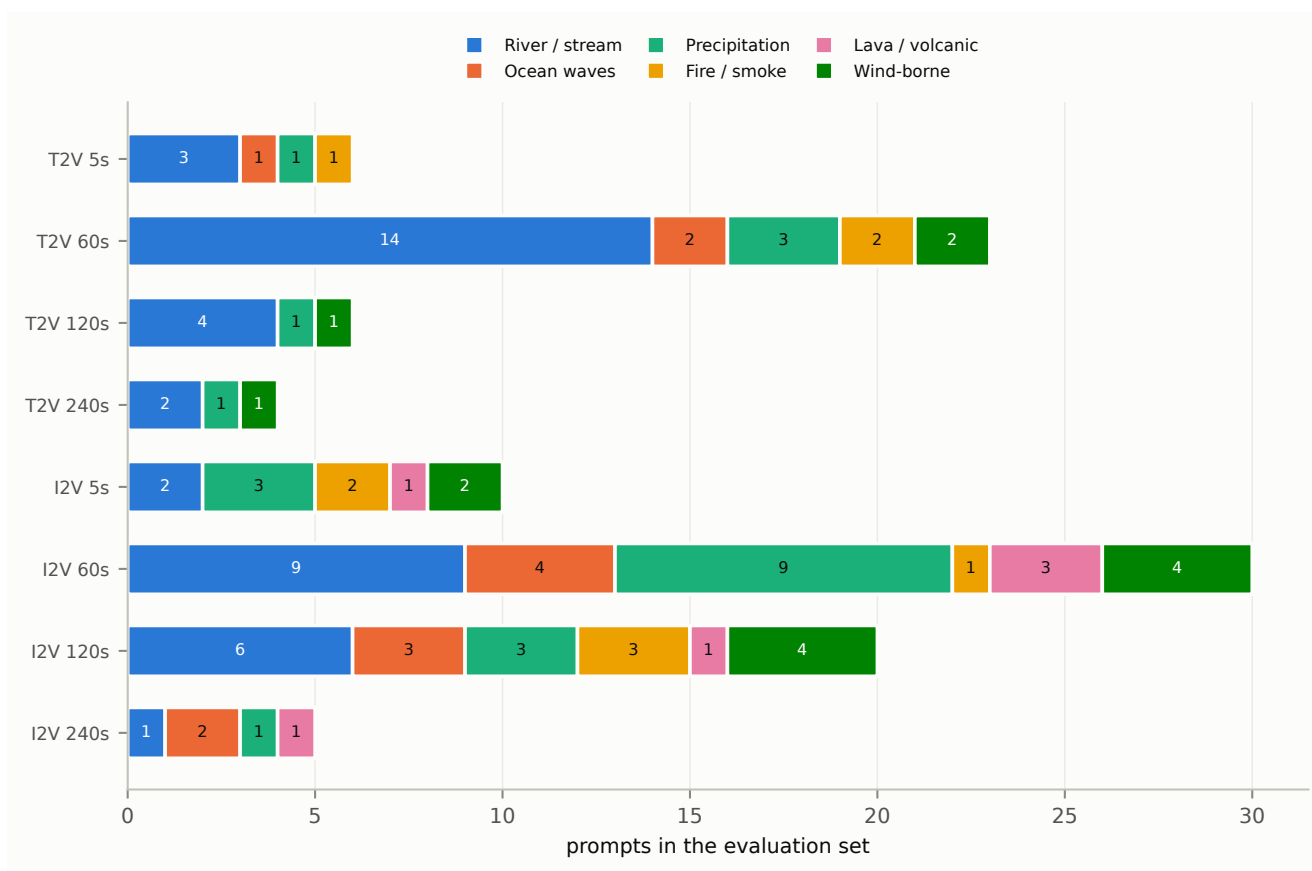


Figure 8. Composition of the evaluation set by scene category, in prompts per category at each track and horizon.

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